



直线相关练习题试卷

一、选择题(每题只有一个正确答案)

1. 直线 $y = 5$ 的倾斜角是()

(The inclination angle of the line $y = 5$ is ())

A. 0° B. $\frac{\pi}{2}$ C. π D. 不存在(no slope)

2. 经过点 $A(0,1)$ 和 $B(1,0)$ 的直线的斜率 k 是()

(The slope k of the line passing through $A(0,1)$ and $B(1,0)$ is ())

A. 1 B. -1 C. 0 D. 2

3. 经过点 $A(2,3)$ 和 $B(-4,6)$ 的直线的斜率 k 是()

(The slope k of the line passing through $A(2,3)$ and $B(-4,6)$ is ())

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $-\frac{1}{3}$ D. $\frac{1}{4}$

4. 直线 $y = -3$ 的倾斜角是()

(The inclination angle of the line $y = -3$ is ())

A. 0° B. $\frac{\pi}{2}$ C. π D. 不存在(no slope)

5. 经过点 $A(0,4)$ 和 $B(3,1)$ 的直线斜率 k 是()

(The slope k of the line passing through $A(0,4)$ and $B(3,1)$ is ())

A. 1 B. -1 C. 0 D. 3

6. 经过点 $A(-1,3)$ 和 $B(5,6)$ 的直线斜率 k 是()

(The slope k of the line passing through $A(-1,3)$ and $B(5,6)$ is ())

A. $\frac{1}{2}$ B. $-\frac{1}{2}$ C. $\frac{3}{2}$ D. $-\frac{3}{2}$

7. 与直线 $x - 2y - 1 = 0$ 垂直的直线是()

(Which line is perpendicular to the line $x - 2y - 1 = 0$?)

A. $2x + y = 0$ B. $x + 2y = 0$ C. $y - 2x = 0$ D. $2y - x = 0$

8. 直线 $x + 4y + 2 = 0$ 和 $6x + 24y - 5 = 0$ 的关系是()

(The relationship between lines $x + 4y + 2 = 0$ and $6x + 12y - 5 = 0$ is ())

A. 平行 (parallel) B. 相交 (intersecting) C. 重合 (coincident) D. 垂直 (perpendicular)

9. 直线 $x - 2y = -4$ 和 $2x - 4y + 8 = 0$ 的关系是()

(The relationship between lines $x - 2y = -4$ and $2x - 4y + 8 = 0$ is ())

A. 平行 (parallel) B. 相交 (intersecting) C. 重合 (coincident) D. 垂直 (perpendicular)

10. 直线 $2x - ay = 3$ 与直线 $x + 3y = 1$ 平行, 那么 a 等于()

(Line $2x - ay = 3$ is parallel to line $x + 3y = 1$, then $a =$ ())

A. $\frac{1}{2}$ B. -6 C. $\pm \frac{1}{2}$ D. $-\frac{1}{6}$

11. 与直线 $2x - y + 3 = 0$ 垂直的直线是()

(Which line is perpendicular to the line $2x - y + 3 = 0$?)

A. $x + 2y = 0$ B. $2x + y = 0$ C. $y - 2x = 0$ D. $2y - x = 0$

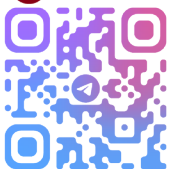
12. 直线 $3x + 6y + 2 = 0$ 和 $6x + 12y - 5 = 0$ 的关系是()

(The relationship between lines $3x + 6y + 2 = 0$ and $6x + 12y - 5 = 0$ is ())

A. 平行 (parallel) B. 相交 (intersecting)
C. 重合 (coincident) D. 垂直 (perpendicular)

13. 直线 $l_1: 2x - 3ay = 4$ 和 $l_2: 4x + y = 2$ 平行, 则实数 a 的值是()

(Line $l_1: 2x - 3ay = 4$ is parallel to $l_2: 4x + y = 2$, then real number $a =$



()

A. $-\frac{1}{6}$ B. $\frac{1}{6}$ C. -6 D. 6

14. 直线 $l_1: 3x + ay = 2$ 和 $l_2: 6x - 4y = 3$ 垂直, 则实数 a 的值是()

(Line $l_1: 3x + ay = 2$ is perpendicular to $l_2: 6x - 4y = 3$, then real number

$a =$ ()

A. $\frac{9}{2}$ B. $-\frac{9}{2}$ C. $\frac{3}{2}$ D. $-\frac{3}{2}$

二、填空题(将答案填在横线上)

1. 直线 l 的倾斜角为 120° , 则它的斜率 $k =$ _____;

(If the inclination angle of line l is 120° , then its slope $k =$ _____;)

2. 直线 l 的倾斜角为 $\frac{5\pi}{6}$, 则它的斜率 $k =$ _____;

(If the inclination angle of line l is $\frac{5\pi}{6}$, then its slope $k =$ _____;)

3. 直线 l 的斜率 $k = -1$, 则倾斜角为 _____;

(If the slope of line l is $k = -1$, then its inclination angle $\alpha =$ _____;)

4. 直线 l 经过点 $(-1, 0)$ 和 $(1, 2)$, 则斜率 $k =$ _____, 倾斜角 _____;

(Line l passes through $(-1, 0)$ and $(1, 2)$, then its slope $k =$ _____, inclination angle $\alpha =$ _____;)

5. 经过点 $A(1, 5)$ 和 $B(-2, 3)$ 的直线的斜率 $k =$ _____;

(The slope k of the line passing through $A(1, 5)$ and $B(-2, 3)$ is _____;)

6. 经过点 $A(-1, x)$ 和 $B(x, 2)$ 的直线的斜率 $k = \frac{2}{3}$, 那么 $x =$ _____;

(The slope of the line passing through $A(-1, x)$ and $B(x, 2)$ is $k = \frac{2}{3}$, then $x =$ _____;)

7. 直线 l 的倾斜角为 135° , 则斜率 $k =$ _____;

(If the inclination angle of line l is 135° , then its slope $k =$ _____;)

8. 直线 l 的倾斜角为 $\frac{2\pi}{3}$, 则斜率 $k =$ _____;

(If the inclination angle of line l is $\frac{2\pi}{3}$, then its slope $k =$ _____;)

9. 直线 l 的斜率 $k = \sqrt{3}$, 则倾斜角 $\alpha =$ _____;

(If the slope of line l is $k = \sqrt{3}$, then its inclination angle $\alpha =$ _____;)

10. 直线 l 经过点 $(-2,1)$ 和 $(3,6)$, 则斜率 $k =$ _____, 倾斜角 $\alpha =$ _____;

(Line l passes through $(-2,1)$ and $(3,6)$, then its slope $k =$ _____, inclination angle $\alpha =$ _____;)

11. 经过点 $A(3,7)$ 和 $B(-1,5)$ 的直线斜率 $k =$ _____;

(The slope k of the line passing through $A(3,7)$ and $B(-1,5)$ is _____;)

12. 经过点 $A(2,x)$ 和 $B(x,6)$ 的直线斜率 $k = \frac{3}{4}$, 则 $x =$ _____;

(The slope of the line passing through $A(2,x)$ and $B(x,6)$ is $k = \frac{3}{4}$, then $x =$ _____;)

13. 直线 $x - 3y + 1 = 0$ 在 x 轴上的截距是_____; 在 y 轴上的截距是_____;

(The x -intercept of the line $x - 3y + 1 = 0$ is _____; the y -intercept is _____;)

14. 直线 $2x - y - 4 = 0$ 的斜率是_____;

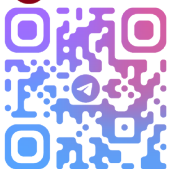
(The slope of the line $2x - y - 4 = 0$ is _____;)

15. 直线 $\frac{x}{5} + \frac{y}{2} = 1$ 的斜率是_____;

(The slope of the line $\frac{x}{5} + \frac{y}{2} = 1$ is _____;)

16. 直线 $y = -6x + 7$ 的斜率是_____;

(The slope of the line $y = -6x + 7$ is _____;)



17. 已知直线 l 的倾斜角是 45° ，经过点 $(3, -5)$ ，那么直线 l 的方程是_____；

(Given that the inclination angle of line l is 45° and it passes through $(3, -5)$, the equation of line l is _____;)

18. 直线 $2x - 5y + 10 = 0$ 在 x 轴上的截距是_____；在 y 轴上的截距是_____；

(The x -intercept of the line $2x - 5y + 10 = 0$ is _____; the y -intercept is _____;)

19. 直线 $3x - 2y - 6 = 0$ 的斜率 $k =$ _____；

(The slope k of the line $3x - 2y - 6 = 0$ is _____;)

20. 直线 $\frac{x}{4} + \frac{y}{-3} = 1$ 的斜率 $k =$ _____；

(The slope k of the line $\frac{x}{4} + \frac{y}{-3} = 1$ is _____;)

21. 直线 $y = 5x - 9$ 的斜率 $k =$ _____；

(The slope k of the line $y = 5x - 9$ is _____;)

22. 已知直线 l 的倾斜角是 60° ，经过点 $(2, -3)$ ，则直线 l 的方程是_____；

(Given that the inclination angle of line l is 60° and it passes through $(2, -3)$, the equation of line l is _____;)

23. 直线 $2x - y + 5 = 0$ 的斜率 $k =$ _____；

(The slope k of the line $2x - y + 5 = 0$ is _____;)

24. 经过点 $P(3, 2)$ ，倾斜角为 60° 的直线方程是_____；

(The equation of the line passing through $P(3, 2)$ with inclination angle 60° is _____;)

25. 斜率为 -4 ，经过点 $(1, -5)$ 的直线方程是_____；

(The equation of the line with slope -4 passing through $(1, -5)$ is _____;)

_____);

26. 直线经过点(4,7)和(0,3), 则斜率 $k =$ _____, 直线方程是_____);

(Line passes through (4,7) and (0,3), then slope $k =$ _____, equation is _____);

27. 在 x 轴和 y 轴的截距分别是-2和6的直线方程是_____;

(The equation of the line with x -intercept -2 and y -intercept 6 is _____);

28. 与直线 $3x + 2y - 1 = 0$ 垂直的直线的斜率是_____;

(The slope of the line perpendicular to $3x + 2y - 1 = 0$ is _____);

29. 直线 $x - y + 5 = 0$ 和 $3x + y + 1 = 0$ 的交点坐标是_____;

(The intersection coordinate of lines $x - y + 5 = 0$ and $3x + y + 1 = 0$ is _____);

30. 直线 $2x - y + 8 = 0$ 和 $x + y + 1 = 0$ 的交点坐标是_____;

(The intersection coordinate of lines $2x - y + 8 = 0$ and $x + y + 1 = 0$ is _____);

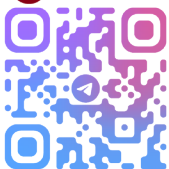
31. 直线 $x + 3y - 7 = 0$ 和 $2x - y - 7 = 0$ 的交点坐标是_____;

(The intersection coordinate of lines $x + 3y - 7 = 0$ and $2x - y - 7 = 0$ is _____);

32. 直线 $x - 2y + 4 = 0$ 和 $2x + y - 1 = 0$ 的交点坐标是_____;

(The intersection coordinate of lines $x - 2y + 4 = 0$ and $2x + y - 1 = 0$ is _____);

33. 直线 $3x + y - 5 = 0$ 和 $x - 3y + 7 = 0$ 的交点坐标是_____;



(The intersection coordinate of lines $3x + y - 5 = 0$ and $x - 3y + 7 = 0$ is _____;)

34. 直线 $2x + 3y - 10 = 0$ 和 $x - y = 0$ 的交点坐标是_____;

(The intersection coordinate of lines $2x + 3y - 10 = 0$ and $x - y = 0$ is _____;)

三、计算题(写出必要的解题步骤，最终答案用最简形式表示)

1. 求经过点 $A(2, -1)$ 和 $B(0, 4)$ 的直线方程;

(Find the equation of the line passing through $A(2, -1)$ and $B(0, 4)$;))

2. 求斜率 $k = \frac{1}{2}$, 经过点 $(1, 3)$ 的直线方程;

(Find the equation of the line with slope $k = \frac{1}{2}$ passing through $(1, 3)$;))

3. 求在 x 轴和 y 轴的截距分别是 -1 和 3 的直线方程;

(Find the equation of the line with x -intercept -1 and y -intercept 3 ;))

4. 求经过点 $A(-1,3)$ 和 $B(2,5)$ 的直线方程;

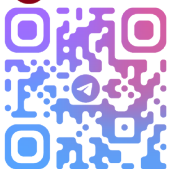
(Find the equation of the line passing through $A(-1,3)$ and $B(2,5)$;))

5. 求经过点 $(3,2)$ 且与 x 轴平行的直线方程;

(Find the equation of the line passing through $(3,2)$ and parallel to the x -axis;))

6. 求经过点 $A(3,-1)$ 和 $B(0,5)$ 的直线方程(用两点式, 再化为一般式);

(Find the equation of the line passing through $A(3,-1)$ and $B(0,5)$ (using two-point form, then convert to general form);))



7. 求斜率 $k = \frac{2}{3}$, 经过点(4,1)的直线方程(用点斜式, 再化为斜截式);

(Find the equation of the line with slope $k = \frac{2}{3}$ passing through (4,1)

(using point-slope form, then convert to slope-intercept form);)

8. 求在 x 轴和 y 轴的截距分别是2和-6的直线方程(用截距式, 再化为一般式);

(Find the equation of the line with x -intercept 2 and y -intercept -6

(using intercept form, then convert to general form);)

9. 求经过点 $A(-2,4)$ 和 $B(3,-1)$ 的直线方程(用任意形式, 化为一般式);

(Find the equation of the line passing through $A(-2,4)$ and $B(3,-1)$

(using any form, then convert to general form);)

10. 求经过点 $(5, -3)$ 且与 x 轴平行的直线方程;

(Find the equation of the line passing through $(5, -3)$ and parallel to the x -axis;)

11. 直线 l_1 经过点 $(-3, 2)$, 且和直线 $l_2: 2x + y + 4 = 0$ 垂直。求: (1) 直线 l_2 的斜率 k_2 ; (2) 直线 l_1 的斜率 k_1 ; (3) 直线 l_1 的方程。

(Line l_1 passes through $(-3, 2)$ and is perpendicular to line $l_2: 2x + y + 4 = 0$. Find: (1) Slope k_2 of line l_2 ; (2) Slope k_1 of line l_1 ; (3) Equation of line l_1 .)

12. 已知直线 $l_1: x - 3y + 10 = 0$, 直线 l_2 经过点 $(2, -1)$, 两条直线互相平行。

求: (1) 直线 l_1 的斜率; (2) 直线 l_2 的斜率; (3) 直线 l_2 的方程。

(Given line $l_1: x - 3y + 10 = 0$, line l_2 passes through $(2, -1)$ and is parallel to l_1 . Find: (1) Slope of line l_1 ; (2) Slope of line l_2 ; (3) Equation of line l_2 .)



13. 直线 l_1 经过点 $(5,4)$, 且和直线 $l_2: 2x - 4y + 3 = 0$ 垂直。求: (1) 直线 l_2 的斜率; (2) 直线 l_1 的斜率; (3) 直线 l_1 的方程。

(Line l_1 passes through $(5,4)$ and is perpendicular to line $l_2: 2x - 4y + 3 = 0$. Find: (1) Slope of line l_2 ; (2) Slope of line l_1 ; (3) Equation of line l_1 .)

14. 直线 l_1 经过点 $(-2,4)$, 且和直线 $l_2: 3x + y + 5 = 0$ 垂直。求: (1) 直线 l_2 的斜率 k_2 ; (2) 直线 l_1 的斜率 k_1 ; (3) 直线 l_1 的方程。

(Line l_1 passes through $(-2,4)$ and is perpendicular to line $l_2: 3x + y + 5 = 0$. Find: (1) Slope k_2 of line l_2 ; (2) Slope k_1 of line l_1 ; (3) Equation of line l_1 .)

15. 已知直线 $l_1: 2x - 3y + 6 = 0$, 直线 l_2 经过点 $(3,2)$, 且与 l_1 平行。求: (1) 直线 l_1 的斜率; (2) 直线 l_2 的斜率; (3) 直线 l_2 的方程。

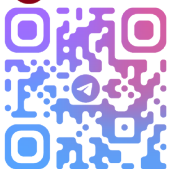
(Given line $l_1: 2x - 3y + 6 = 0$, line l_2 passes through $(3,2)$ and is parallel to l_1 . Find: (1) Slope of line l_1 ; (2) Slope of line l_2 ; (3) Equation of line l_2 .)

16. 直线 l_1 经过点 $(4,5)$, 且和直线 $l_2: 6x - 3y + 2 = 0$ 垂直。求: (1) 直线 l_2 的斜率; (2) 直线 l_1 的斜率; (3) 直线 l_1 的方程。

(Line l_1 passes through $(4,5)$ and is perpendicular to line $l_2: 6x - 3y + 2 = 0$. Find: (1) Slope of line l_2 ; (2) Slope of line l_1 ; (3) Equation of line l_1 .)

17. 求经过 $x + 2y - 7 = 0$ 和 $3x - y = 0$ 的交点, 且平行于直线 $l_1: 2x + y + 5 = 0$ 的直线 l_2 的方程;

(Find the equation of line l_2 that passes through the intersection of $x + 2y - 7 = 0$ and $3x - y = 0$, and is parallel to line $l_1: 2x + y + 5 = 0$.)



18. 已知直线 l_1 经过 $l_2: 2x - 3y + 5 = 0$ 和 $l_3: 2x + 5y - 11 = 0$ 的交点, 且与直线 l_2 垂直。求: (1) 直线 l_2 和 l_3 的交点坐标; (2) 直线 l_1 的斜率; (3) 直线 l_1 的方程。

(Given that line l_1 passes through the intersection of $l_2: 2x - 3y + 5 = 0$ and $l_3: 2x + 5y - 11 = 0$, and is perpendicular to l_2 . Find: (1) Intersection coordinate of l_2 and l_3 ; (2) Slope of l_1 ; (3) Equation of l_1 .)

19. 求经过直线 $3x - 2y + 8 = 0$ 和 $x + 3y - 1 = 0$ 的交点, 且与直线 $x - y + 2 = 0$ 平行的直线方程;

(Find the equation of the line passing through the intersection of $3x - 2y + 8 = 0$ and $x + 3y - 1 = 0$, and parallel to $x - y + 2 = 0$;))

20. 求经过点 $A(3, -4)$ 且与直线 $4x - 3y + 2 = 0$ 垂直的直线方程;

(Find the equation of the line passing through $A(3, -4)$ and perpendicular to $4x - 3y + 2 = 0$;))

21. 已知直线 l 经过点 $(2,1)$, 且与直线 $l_1: x + 2y - 3 = 0$ 平行, 求直线 l 与 l_1 之间的距离;

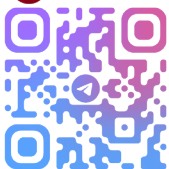
(Given line l passes through $(2,1)$ and is parallel to $l_1: x + 2y - 3 = 0$, find the distance between l and l_1 .)

22. 求经过点 $P(1,3)$, 且与 x 轴、 y 轴的正半轴分别交于 A 、 B 两点, 使得 $\triangle AOB$ (O 为原点) 的面积为 6 的直线方程。

(Find the equation of the line passing through $P(1,3)$ and intersecting the positive semi-axes of x -axis and y -axis at A and B respectively, such that the area of $\triangle AOB$ (O is the origin) is 6.)

23. 求经过直线 $x + 3y - 8 = 0$ 和 $2x - y - 3 = 0$ 的交点, 且平行于直线 $l_1: 3x + y + 4 = 0$ 的直线 l_2 的方程;

(Find the equation of line l_2 that passes through the intersection of $x + 3y - 8 = 0$ and $2x - y - 3 = 0$, and is parallel to line $l_1: 3x + y + 4 = 0$.)



24. 已知直线 l_1 经过 $l_2: 3x - 2y + 4 = 0$ 和 $l_3: 4x + y - 9 = 0$ 的交点, 且与直线 l_2 垂直。求: (1) 直线 l_2 和 l_3 的交点坐标; (2) 直线 l_1 的斜率; (3) 直线 l_1 的方程。

(Given that line l_1 passes through the intersection of $l_2: 3x - 2y + 4 = 0$ and $l_3: 4x + y - 9 = 0$, and is perpendicular to l_2 . Find: (1) Intersection coordinate of l_2 and l_3 ; (2) Slope of l_1 ; (3) Equation of l_1 .)